
a Estimates under each robustness check (main estimate beneath each heading; † p ≥ 0.05)
E-scooter vs. bicycle +2.1
E-bike vs. bicycle +5.1
Age 65+ vs. 25–64 +12.9
Male vs. female +2.0
≤30 vs. 50 km/h −2.0
Cycle facility −2.2
Heavy vehicle +17.6
Single vehicle +12.5
Cycle path −4.5
Including 2019 +2.7 +5.1 +13.3 +2.1 −1.8 −2.0 +18.5 +11.6 −4.4
Excluding the Paris police area +2.7 +6.7 +18.2 +2.5 −3.0 −3.1 +19.3 +15.2 −6.3
Fatal outcome +0.8† +1.0† +5.9 +0.9 −0.3† −0.4† +9.7 +4.5 −1.6
Motor-vehicle collisions only +0.6† +4.5 +12.3 +1.8 −1.8 −0.9† +17.6 – −5.0
Early-adopter départements +0.9† – – – – – – – –
Any-row location variables +2.0 +5.1 +12.9 +2.0 −2.0 −2.2 +17.5 +12.5 −4.7
Without location-file variables +0.8† +5.2 +12.9 +2.0 −3.1 −3.3 +18.2 +14.4 −3.9
Random-forest nuisance models +1.5 +3.9 +15.0 +2.6 −2.8 −3.8 +17.5 +9.6 −5.0
Logistic nuisance models +1.2 +4.4 +13.4 +2.9 −1.6 −3.2 +17.7 +11.5 −4.7
Trimming at 0.01 +2.3 +5.3 +12.9 +2.0 −2.0 −2.1 +17.3 +12.3 −4.2
Trimming at 0.05 +1.8 +4.7 +13.1 +2.0 −2.0 −2.4 +18.2 +12.4 −4.5
Partially linear model +2.4 +3.4 +15.8 +2.0 −2.4 −1.7 +16.7 +11.3 −2.2
Ratio to the main estimate
2.0
1.5
1.0
0.5
0.0
b E-scooter vs. bicycle under each check
Main analysis +2.1 (0.8 to 3.3)
Including 2019 +2.7 (1.4 to 3.9)
Excluding the Paris police area +2.7 (0.9 to 4.5)
Fatal outcome +0.8 (−0.1 to 1.7)
Motor-vehicle collisions only +0.6 (−0.8 to 2.0)
Early-adopter départements +0.9 (−0.4 to 2.2)
Any-row location variables +2.0 (0.8 to 3.3)
Without location-file variables +0.8 (−0.4 to 2.1)
Random-forest nuisance models +1.5 (0.5 to 2.5)
Logistic nuisance models +1.2 (0.3 to 2.2)
Trimming at 0.01 +2.3 (0.9 to 3.6)
Trimming at 0.05 +1.8 (0.7 to 2.9)
Partially linear model +2.4 (1.6 to 3.3)
−2
0
2
4
Double ML estimate (points, 95% CI)
c By recording context
All départements (n = 42,994)
−9.0
+2.1
Ten early-adopter départements (n = 20,214)
−0.1
+0.9
All other départements (n = 22,780)
−14.1
+3.1
Unadjusted
Double ML (95% CI)
−10
0
Difference (points)
d Sensitivity to unobserved confounding (RV, robustness value)
Absolute main estimate
Change produced by an unobserved confounder as strong as département
Percentage points
25
20
15
10
5
0
RV 1.4%
2.1
0.5
RV 3.4%
5.1
0.9
RV 7.1%
12.9
8.8
RV 2.1%
2.0
1.5
RV 1.9%
2.0
3.8
RV 1.7%
2.2
2.0
RV 8.7%
17.6
0.4
RV 6.0%
12.5
6.2
RV 1.8%
4.5
1.0
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